

Auteur: Nathan De Roy  
 Studierichting: Informatica  
 Oefeningenreeks 3H nr. 12.1

**Schets de functie:**  $r = 3(1 - \sin(\theta))$

$$\left. \begin{array}{l} \text{dom } r = \mathbb{R} \\ P = 2\pi \end{array} \right\} \text{Bep. Dom.} = [0, 2\pi]$$

**Nulpunten:**

$$r = 0 \Leftrightarrow \sin(\theta) = 0 \Leftrightarrow \theta = \frac{\pi}{2} + k2\pi \text{ met } k \in \mathbb{N} \longrightarrow \theta \in \left\{ \frac{\pi}{2} \right\}$$

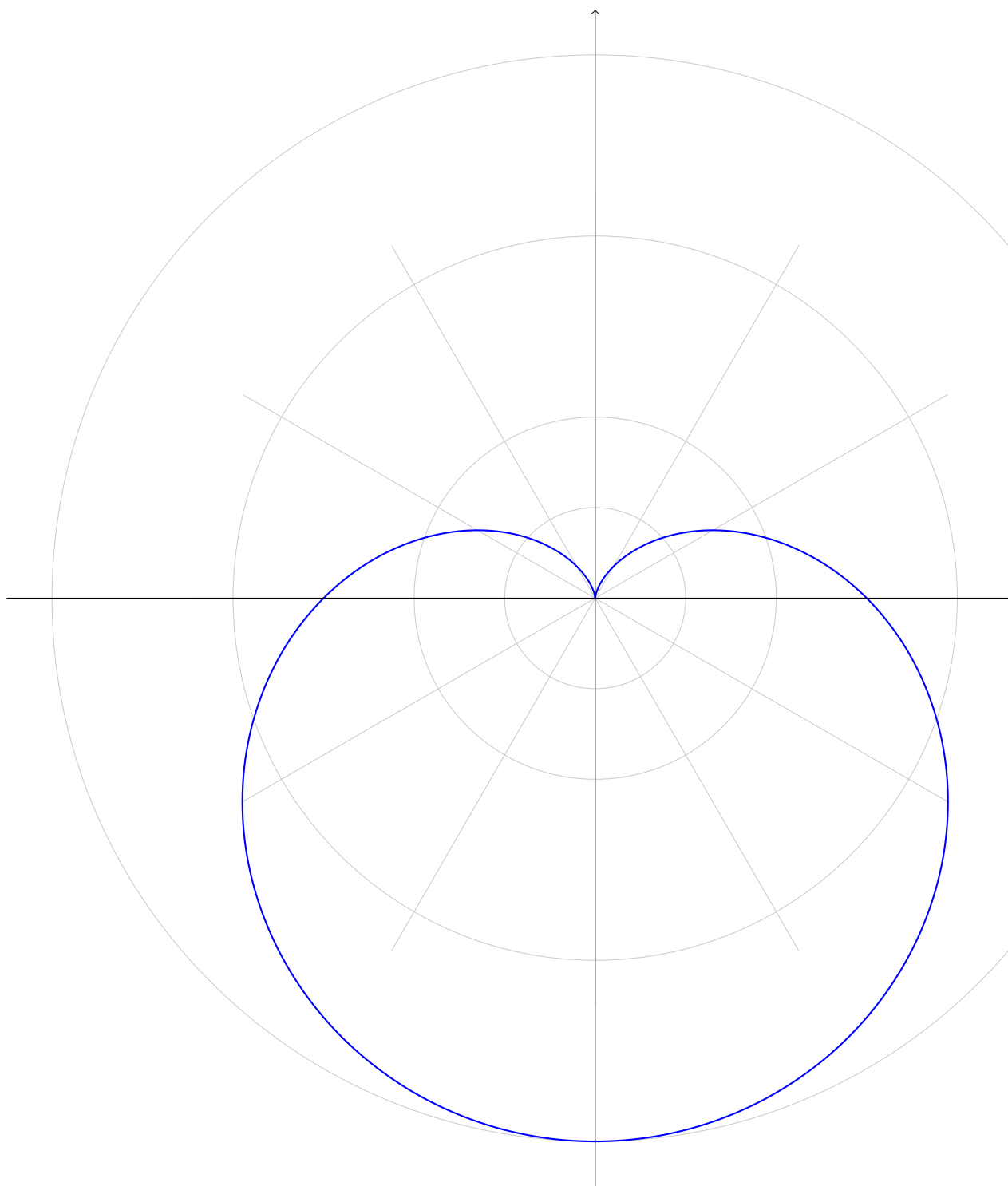
$$r' = 0 \Leftrightarrow -3 \cos(\theta) = 0 \Leftrightarrow \cos(\theta) = 0 \Leftrightarrow \theta = \frac{\pi}{2} + k\pi \text{ met } k \in \mathbb{N}$$

$$\longrightarrow \theta \in \left\{ \frac{\pi}{4}, \frac{3\pi}{4} \right\}$$

**Schema:**

| $\theta$ | 0          |            | $\frac{\pi}{2}$ |            | $\pi$      |            | $\frac{3\pi}{2}$ |            | $2\pi$     |  |
|----------|------------|------------|-----------------|------------|------------|------------|------------------|------------|------------|--|
| $r$      | 3          | +          | 0               | +          | 3          | +          | 6                | +          | 3          |  |
| $r'$     | -3         | -          | 0               | +          | 3          | +          | 0                | -          | -3         |  |
|          | $\searrow$ | $\searrow$ | min             | $\nearrow$ | $\nearrow$ | $\nearrow$ | Max              | $\searrow$ | $\searrow$ |  |

**Grafiek in poolcoördinaten:**



## References

[1]